BACHELOR IN SOFTWARE ENGINEERING (HONOURS)

FINAL EXAMINATION
JANUARY 2022

Course : EC3302 (Theory of Computation) — Time: 1.30pm – 12.30pm (3 hours)

Lecturer : Ms. Rajesvary Rajoo — Date: 22 April 2022

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **4** printed pages (excluding front cover).

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Consider the following code:

```
class Cat {
        public void move(){
                System.out.println("Walk");
        }
 // other methods
}
```

i Adapt this code so that the Strategy pattern is used to make each different type of Animal move in a different way. (Note: You may use class diagram or code to answer this question.) **(6 MARKS)**

ii If the program needed a cat that move, "Jump" instead of "Walk" describe how would you create one? **(4 MARKS)**

iii Explain any **ONE (1)** of the design principles that is being used in the above code. **(5 MARKS)**

b. Distinguish Abstract Factory and Factory Method. **(4 MARKS)**

c. Give an example situation when you might use a Singleton pattern and explain the major features of the code required. You do not need to write any actual code. **(6 MARKS)**

2 a. Explain by using a suitable example why defensive programming is not encouraged to be used in programming. Suggest an alternative. **(7 MARKS)**

b. Consider the following two method specifications:

Case 1:

```
/*@ requires (\exists int x; ; a == x*b);
```

```
   @ ensures a == \result*b;
   @*/
static int exactDivide1 (int a, int b) { . . . }
```

Case 2 :

```
/*@ ensures a== \result*b;
   @ signals (DivException e) !(\exists in x;  ; a == x*b)
   @*/
static int exactDivide1  (int a, int b) throws DivException { ... }
```

In Case 1, method exactDivide1 has precondition P: $\Leftrightarrow \exists x : a = x.b$ and in Case 2 exactDivide1 has precondition P: true.

i Explain what will happen if P is false in Case 1? **(2 MARKS)**

ii Explain what is meant by the statement of a precondition is true? What will happen if P is false in Case 2? **(4 MARKS)**

c. Explain why preconditions cannot be ANDed together just like post conditions and object invariants in inheritance when subclass inherits specifications from the superclass. **(6 MARKS)**

d. Explain any **ONE (1)** support that Java provides for checking pre-conditions and post-conditions. Illustrate your answer by using suitable example. **(6 MARKS)**

3. a. Identify a technique that would allow us to store objects of different types in a single generic data structure. Illustrate your answer with an example. **(5 MARKS)**

i Rewrite the printList() method using an Iterator. **(5 MARKS)**

ii Explain why using Iterator is a better idea than using the get(index) method. **(3 MARKS)**

c i What purpose do types serve in a programming language? **(3 MARKS)** ii What does it mean for a language to be Statically typed and Dynamically typed? **(4 MARKS)**

iii Explain why generics make more sense as a feature in statically typed language than in dynamically typed language. **(5 MARKS)**

4 a. Multidimensional arrays in C# can be either jagged or rectangular

i Differentiate jagged and rectangular array. **(6 MARKS)**

ii Discuss the relative merits of using jagged arrays over rectangular array **(6 MARKS)**

b. Consider the following statement: **(8 MARKS)**

"Limited reintroduction of the goto in programming is not considered harmful if used wisely."

Do you agree with this statement? Give a simple code example to illustrate your answer. Otherwise, give a counter argument with an example code.

c. Differentiate software component and component software **(5 MARKS)**

-END OF QUESTION PAPER-